

Chapter 07

Multiple Regression Analysis

Dr. NEANG Pheak

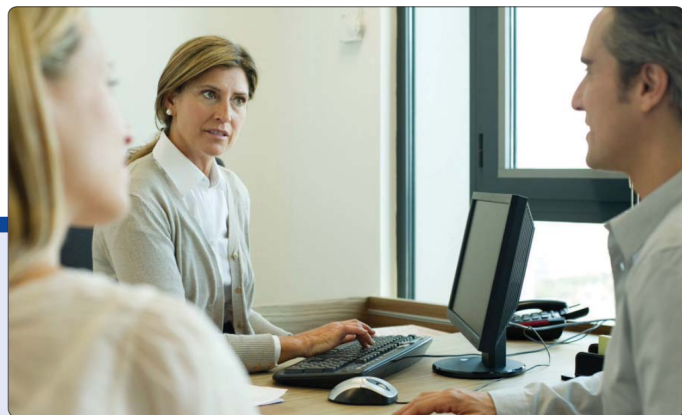
Institute of Technology of Cambodia

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Learning Objectives

When we have completed this chapter, we will be able to:



(1) Use multiple regression analysis to describe and interpret a relationship between several independent variables and a dependent

variable.

- (2) Evaluate how well a multiple regression equation fits the data.
- (3) Test hypotheses about the relationships inferred by a multiple regression model.
- (4) Evaluate the assumptions of multiple regression.
- (5) Use and interpret a qualitative, dummy variable in multiple regression.

What is the multiple linear regression?

We use k to represent the number of independent variables. So k can be any positive integer.

$$\hat{y} = a + b_1x_1 + b_2x_2 + \dots + b_kx_k$$

- a is the y -intercept, the value of $\hat{y}$ when all the x 's are zero.

- b_j is the amount by which $\hat{y}$ changes when that particular x_j increases by one unit, with the values of all other independent variables held constant.

What is the multiple linear regression?

When there are two independent variables, the regression equation is

$$\hat{y} = a + b_1x_1 + b_2x_2$$

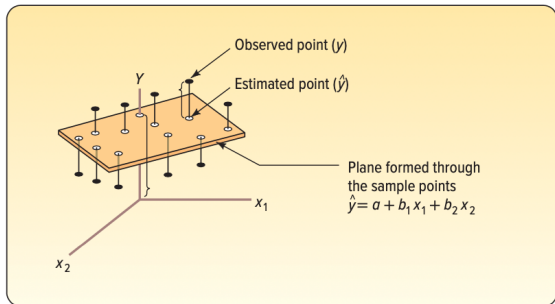


CHART 14-1 Regression Plane with 10 Sample Points

- The chart shows the residuals as the difference between the actual y and the fitted $\hat{y}$ on the plane.
- Let x_1 refer to the number of rooms, x_2 to the age of the home in years, and y to the selling price of the home in thousands of dollars (\$000).

Example: Salsberry Realty sells homes

TABLE 14-1 Factors in January Heating Cost for a Sample of 20 Homes

Home	Heating Cost (\$)	Mean Outside Temperature (°F)	Attic Insulation (inches)	Age of Furnace (years)
1	\$250	35	3	6
2	360	29	4	10
3	165	36	7	3
4	43	60	6	9
5	92	65	5	6
6	200	30	5	5
7	355	10	6	7
8	290	7	10	10
9	230	21	9	11
10	120	55	2	5
11	73	54	12	4
12	205	48	5	1
13	400	20	5	15
14	320	39	4	7
15	72	60	8	6
16	272	20	5	8
17	94	58	7	3
18	190	40	8	11
19	235	27	9	8
20	139	30	7	5

- (1) Determine the multiple regression equation. Which variables are the independent variables? Which variable is

the dependent variable?

Discuss the regression coefficients.

- (2) What does it indicate if some coefficients are positive and some coefficients are negative? What is the intercept value? What is the estimated heating cost for a home if the mean outside temperature is 30 degrees, there are 5 inches of insulation in the attic, and the furnace is 10 years old?

Solution: Salsberry Realty sells homes

The estimated regression equation is:

$$\hat{y} = 427.194 - 4.583x_1 - 14.831x_2 + 6.101x_3$$

- The mean outside temperature for the month is $\hat{y} = 427.194 - 4.583(30) - 14.831(5) + 6.101(10)$
- The estimated January heating cost is \$276.56.

#	A	B	C	D	E	F	G	H	I	J	K
1	Cost	Temp	Insul	Age		SUMMARY OUTPUT					
2	250	35	3	6							
3	360	29	4	10		Regression Statistics					
4	165	36	7	3		Multiple R	0.897				
5	43	60	6	9		R Square	0.804				
6	92	65	5	6		Adjusted R Square	0.767				
7	200	30	5	5		Standard Error	51.049				
8	355	10	6	7		Observations	20				
9	290	7	10	10							
10	230	21	9	11		ANOVA					
11	120	55	2	5			df	SS	MS	F	Significance F
12	73	54	12	4		Regression	3	171220.473	57073.491	21.901	0.000
13	205	48	5	1		Residual	16	41695.277	2605.955		
14	400	20	5	15		Total	19	212915.750			
15	320	39	4	7							
16	72	60	8	6			Coefficients	Standard Error	t Stat	P-value	
17	272	20	5	8		Intercept	427.194	59.601	7.168	0.000	
18	94	58	7	3		Temp	-4.583	0.772	-5.934	0.000	
19	190	40	8	11		Insul	-14.831	4.754	-3.119	0.007	
20	235	27	9	8		Age	6.101	4.012	1.521	0.148	
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Source: Microsoft Excel

Evaluating a multiple regression equation

Using the information in an ANOVA table to evaluate how well the equation fits the data.

Source	<i>df</i>	SS	MS	<i>F</i>
Regression	k	SSR	$MSR = SSR/k$	MSR/MSE
Residual or error	$\frac{n - (k + 1)}{n - 1}$	SSE	$MSE = SSE/[n - (k + 1)]$	
Total		SS total		

(1) The total sum of squares = SS Total = $\sum (y - \hat{y})^2$

(2) Regression Sum of Squares = SSR = $\sum (\hat{y} - \bar{y})^2$

(3) Residual or Error Sum of Squares = SSE = $\sum (y - \hat{y})^2$

To estimate January heating costs

The first sampled home in row 2 in the Excel spreadsheet above.

	A	B	C	D	G	H	I	J	K	L	M
1	Cost	Temp	Insul	Age		SUMMARY OUTPUT					
2	250	35	3	6							
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Source: Microsoft Excel

- (1) The actual heating cost for the first observation, y , is \$250; the outside temperature, x_1 , is 35 degrees; the depth of insulation, x_2 , is 3 inches; and the age of the furnace, x_3 , is 6 years.

To estimate January heating costs

$$\begin{aligned}\hat{y} &= 427.194 - 4.583x_1 - 14.831x_2 + 6.101x_3 \\ &= 427.194 - 4.583(35) - 14.831(3) + 6.101(6) \\ &= 258.90\end{aligned}$$

So we would estimate that a home with a mean January outside temperature of 35 degrees, 3 inches of insulation, and a 6-year-old furnace would cost \$258.90 to heat.

Multiple standard error of estimate

We can calculate the multiple standard error of the estimate as:

$$S_{y \cdot 12 \dots k} = \sqrt{\frac{\sum (y - \hat{y})^2}{n - (k + 1)}} = \sqrt{\frac{\mathbf{SSE}}{n - (k + 1)}} \quad (1)$$

- y is the actual observation.
- $\hat{y}$ is the estimated value computed from the regression equation.
- n is the number of observations in the sample.
- k is the number of independent variables.
- **SSE** is the residual sum of squares from an ANOVA table.

How to interpret the standard error of estimate ?

A smaller multiple standard error indicates a better or more effective predictive equation.

(1) The units are the same as the dependent variable, so the standard error is in dollars, \$51.05.

	A	B	C	D	G	H	I	J	K	L	M
1	Cost	Temp	Insul	Age		SUMMARY OUTPUT					
2	250	35	3	6							
3	360	29	4	10		Regression Statistics					
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(2) We expect the residuals to be approximately normally distributed, so about 68% of the residuals will be within \$51.05 and about 95% within $\pm 2(51.05)$, or $\pm \$102.10$.

Coefficient of Multiple Determination

The percent of variation in the dependent variable, y , explained by the set of independent variables, $x_1, x_2, \dots, x_k$

$$R^2 = \frac{\mathbf{SSR}}{\mathbf{SS\ total}} \quad (2)$$

The characteristics of the coefficient of multiple determination are:

- (1) A value near 0 indicates little association between the set of independent variables and the dependent variable
- (2) A value near 1 means a strong association

How to interpret the standard error of estimate ?

$$R^2 = \frac{\text{SSR}}{\text{SS total}} = \frac{171,220.473}{212,915.750} = .804$$

	A	B	C	D	G	H	I	J	K	L	M
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2	250	35	3	6							
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Source: Microsoft Excel

- (1) We conclude that the independent variables (outside temperature, amount of insulation, and age of furnace) explain, or account for, 80.4% of the variation in heating cost.
- (2) To put it another way, 19.6% of the variation is due to other sources, such as random error or variables not included in the analysis

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- Hence, R^2 increases only because the total number of independent variables increases and not because the added independent variable is a good predictor of the dependent variable
- In fact, if the number of variables, k , and the sample size, n , are equal, the coefficient of determination is 1
- In practice, this situation is rare and would also be ethically questionable. To balance the effect that the number of independent variables has on the coefficient of multiple determination, statistical software packages use an adjusted coefficient of multiple determination

Adjusted Coefficient of Determination

$$R_{adj}^2 = 1 - \frac{\frac{\text{SSE}}{n-(k+1)}}{\frac{\text{SS total}}{n-1}} = 1 - \frac{\frac{41,695.277}{20-(3+1)}}{\frac{212,915.750}{20-1}} = 1 - .233 = .767 \quad (3)$$

If we compare the $R^2(0.80)$ to the adjusted $R^2(0.767)$, the difference in this case is small

Restaurants in Myrtle Beach

Analysis of Variance

Source	DF	SS	MS
Regression	5	100	20
Residual Error	20	40	2
Total	25	140	

- (1) How large was the sample?
- (2) How many independent variables are there?
- (3) How many dependent variables are there?
- (4) Compute the standard error of estimate. About 95% of the residuals will be between what two values?
- (5) Determine the coefficient of multiple determination. Interpret this value.
- (6) Find the coefficient of multiple determination, adjusted for the degrees of freedom.

Inferences in Multiple Linear Regression

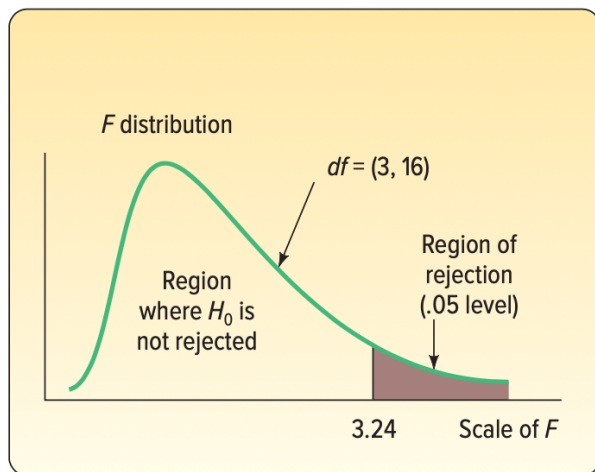
A test used to determine if any of the set of independent variables has regression coefficients different from zero.

β represents the population slope for the regression equation.

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$$

$$H_1 : \text{not all the } \beta_i \text{ are } 0$$

- If the hypothesis test fails to reject the null hypothesis, it implies the regression coefficients are all zero and, logically, are of no value in estimating the dependent variable (heating cost).
- Should that be the case, we would have to search for some other independent variables—or take a different approach—to predict home heating costs.



Adjusted Coefficient of Determination

$$F = \frac{\text{SSR}/k}{\text{SSE}/[n - (k + 1)]} = \frac{171,220.473/3}{41,695.277/[20 - (3 + 1)]} = 21.90$$

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- In our global multiple regression hypothesis test, we will reject the null hypothesis, H_0 , that all regression coefficients are zero when the regression mean square is larger in comparison to the residual mean square.

Evaluating Individual Regression Coefficients

Our strategy is to use three sets of hypotheses: one for temperature, one for insulation, and one for age of the furnace.

For temperature:

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

For insulation:

$$H_0 : \beta_2 = 0$$

$$H_1 : \beta_2 \neq 0$$

For furnace age:

$$H_0 : \beta_3 = 0$$

$$H_1 : \beta_3 \neq 0$$

Evaluating Individual Regression Coefficients

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Source: Microsoft Excel

- (1) $\hat{y} = 427.194 - 4.583x_1 - 14.831x_2 + 6.101x_3$
- (2) Interpreting the term $-4.583x_1$ in the equation: For each degree increase in temperature, we predict that heating cost will decrease \$4.58, holding the insulation and age of the furnace variables constant.

Evaluating Individual Regression Coefficients

	A	B	C	D	G	H	I	J	K	L	M
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Evaluating Individual Regression Coefficients

Testing Individual Regression Coefficient

$$t = \frac{b_i - 0}{S_{b_i}}$$

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- (1) The computed value of t is -5.937 for temperature (the small difference between the computed value and that shown on the Excel output is due to rounding) and -3.119 for insulation.
- (2) Both of these t -values are in the rejection region to the left of -2.120 . Thus, we conclude that the regression coefficients for the temperature and insulation variables are not zero.

Homoscedasticity

The variation around the regression equation is the same for all of the values of the independent variables.

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